Do not write outside the box

0 4 Tap water must be safe to drink.

0 4 . 1 What name is given to water that is safe to drink?

[1 mark]

Tick (✓) **one** box.

Ground water ☐

Potable water ☐

Waste water ☐

0 4 . 2 Water is sterilised to make the water safe to drink.

Which **two** of the following are used to sterilise drinking water?

[2 marks]

Tick (✓) **two** boxes.

Carbon dioxide ☐

Electrolysis ☐

Filtration ☐

Ozone ☐

Ultraviolet light ☐

Question 4 continues on the next page

Do not write outside the box

A student investigated the mass of dissolved solids in samples of river water, sea water and tap water.

This is the method used.

1. Weigh an evaporating basin.

2. Measure 100 cm^3 of river water.

3. Pour the river water into the evaporating basin.

4. Heat the evaporating basin until all the water has evaporated.

5. Weigh the evaporating basin and dissolved solids.

6. Calculate the mass of dissolved solids in the water.

7. Repeat steps 1 to 6 with sea water and then with tap water.

0 4 . 3 Which is the most suitable equipment to measure 100 cm^3 of water?

[1 mark]

Tick (✓) **one** box.

Beaker ☐

Conical flask ☐

Measuring cylinder ☐

Do not write outside the box

0 4 . 4 **Table 3** shows the results.

Table 3

Type of water	Mass in grams		
	Evaporating basin	**Evaporating basin and dissolved solids**	**Dissolved solids**
River	112.1	113.1	1.0
Sea	110.5	114.0	**X**
Tap	115.3	115.4	0.1

Calculate value **X** in **Table 3**.

[1 mark]

X = _____ g

0 4 . 5 Identify the variables used in the investigation.

Draw **one** line from each variable to the example of the variable.

[2 marks]

Variable	**Example of variable**
	Mass of dissolved solids
	Mass of evaporating basin
Control	Room temperature
Dependent	Type of water
	Volume of water

Turn over ►

Do not write outside the box

Table 4 shows the mass of different types of ions dissolved in 1 dm^3 of sea water.

Table 4

Type of ion	**Mass of type of ion dissolved in 1 dm^3 of sea water in grams**
Calcium	0.4
Magnesium	1.3
Sulfate	2.7

Do not write outside the box

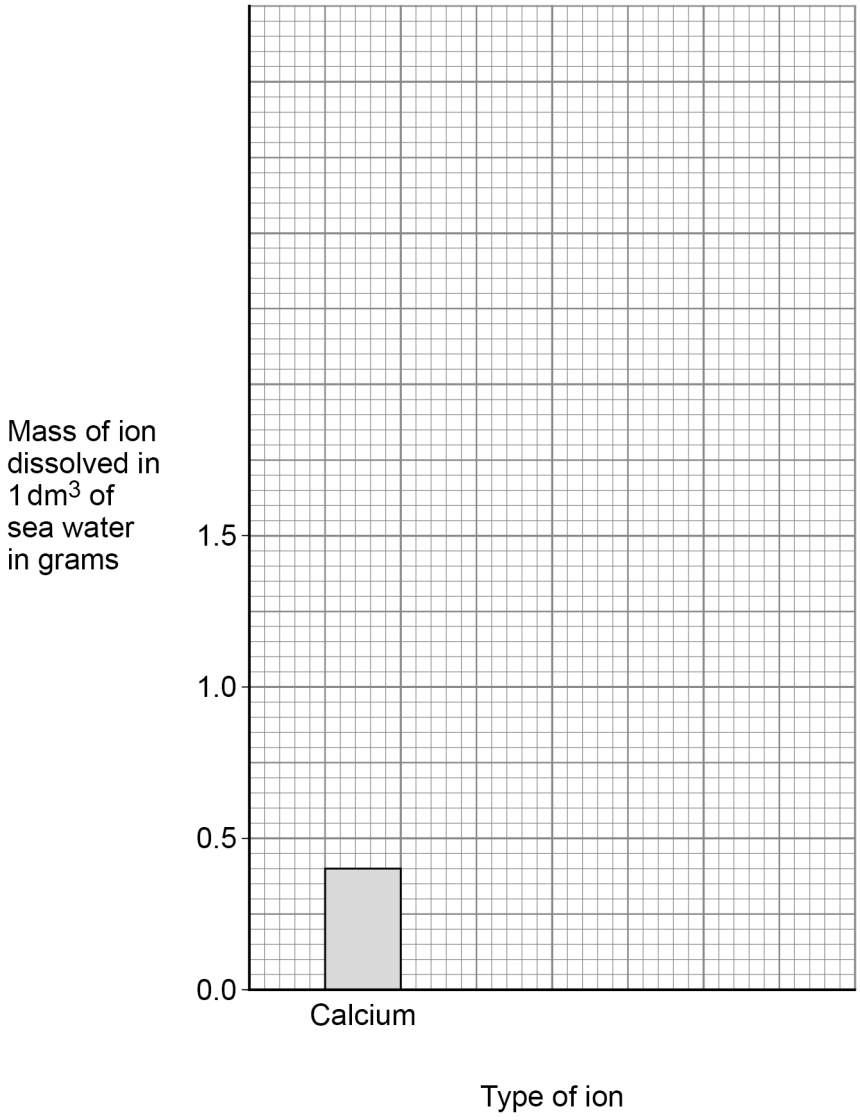
0 4 . 6 Complete **Figure 3**.

You should:

- complete the scale for the *y*-axis
- plot the data from **Table 4** as a bar chart.

[3 marks]

Figure 3



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Turn over for the next question